ZONE 02 OF 07 · GAS TURBINES, GENERATORS, STEAM TURBINE

POWER ISLAND

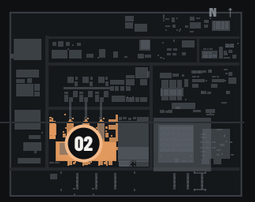
Three H-class gas turbines and a steam turbine share one hall. Every auxiliary that keeps them turning, from lube oil to fuel valves to the static starters, is fed and controlled by cable.

ARMOR-X MC-HL 15 kV (red) and MV-105 in the tray over GT1, control cable in the tier below: the legend on every jacket.



THE TOP FIVE

Numbers on the view mark where each product runs. Cross-sections: typical construction of each cable family.





ARMOR-X® MC-HL 15 KV, RED PVC

13.8 kV feeders to auxiliary drives through the Class I, Div. 2 turbine areas, laid in tray with no conduit.



ARMOR-X® MC-HL 600 V, BLACK PVC

Motor and power circuits on the turbine enclosures, lube-oil and fuel skids.



MV-105 15 KV, EPR 133%

Switchgear to unit transformers, boiler feed pumps and the static starters.



VFD CABLE, SYMMETRICAL GROUNDS, SHIELDED

Drives to pumps and fans: controls noise and reflected-wave stress.



INSTRUMENTATION CABLE (PLTC / ITC)

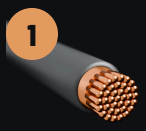
Turbine and HRSG control loops, transmitters and thermocouples.

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06 BATTERY STORAGE

The battery yard pairs containers with power-conversion skids. DC conductors run in precast trenches; 35 kV collection loops carry the power to the yard substation. Paying 1,500 V DC conductor off the reel into the precast trench between the battery containers.



PV / RHW-2 2 KV DC CONDUCTOR

Battery racks to inverters in precast trenches (1,500 V systems).



MV-105 35 KV ALUMINIUM, CONCENTRIC NEUTRAL

Collection loops from the PCS / MV skids to the collector e-house.



TYPE TC-ER 600 V

Container HVAC, auxiliary power and the energy management system.



SHIELDED CONTROL / COMMUNICATIONS CABLE

Battery management, inverter and SCADA links.



BARE COPPER GROUNDING

Container and skid grounding to the yard grid.

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